

## SAT 7

## Math Module 1

Question # ID  
SAT7 M 1.1 a07ma101

Answer

A

Lorenzo purchased a box of cereal and some strawberries at the grocery store. Lorenzo paid \$2 for the box of cereal and \$1.90 per pound for the strawberries. If Lorenzo paid a total of \$9.60 for the box of cereal and the strawberries, which of the following equations can be used to find  $p$ , the number of pounds of strawberries Lorenzo purchased? (Assume there is no sales tax.)

(A)  $1.90p + 2 = 9.60$

(B)  $1.90p - 2 = 9.60$

(C)  $1.90 + 2p = 9.60$

(D)  $1.90 - 2p = 9.60$

SAT7 M 1.2 a07ma102

C

The function  $f$  is defined by  $f(x) = 25x + 30$ . What is the value of  $f(x)$  when  $x = 2$ ?

(A) 50

(B) 57

(C) 80

(D) 110

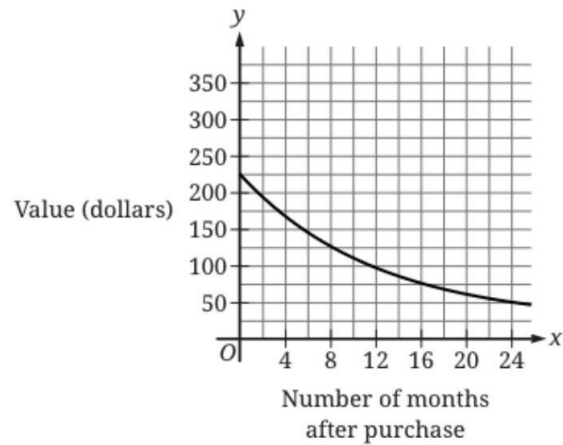
# SAT 7

## Math Module 1

Question # ID  
SAT7 M 1.3 a07ma103

Answer

A



The graph shown gives the estimated value, in dollars, of a tablet as a function of the number of months since it was purchased. What is the best interpretation of the y-intercept of the graph in this context?

- (A) The estimated value of the tablet was \$225 when it was purchased.
- (B) The estimated value of the tablet 24 months after it was purchased was \$225.
- (C) The estimated value of the tablet had decreased by \$225 in the 24 months after it was purchased.
- (D) The estimated value of the tablet decreased by approximately 2.25% each year after it was purchased.

SAT7 M 1.4 a07ma104

$$|p| + 61 = 65$$

B

Which value is a solution to the given equation?

- (A)  $\frac{65}{61}$
- (B) 4
- (C) 126
- (D) 130

## SAT 7

## Math Module 1

Question # ID  
SAT7 M 1.5 a07ma105

Answer

B

Triangles  $EFG$  and  $JKL$  are congruent, where  $E$ ,  $F$ , and  $G$  correspond to  $J$ ,  $K$ , and  $L$ , respectively. The measure of angle  $E$  is  $45^\circ$  and the measure of angle  $F$  is  $20^\circ$ . What is the measure of angle  $J$ ?

(A)  $20^\circ$

(B)  $45^\circ$

(C)  $135^\circ$

(D)  $160^\circ$

# SAT 7

## Math Module 1

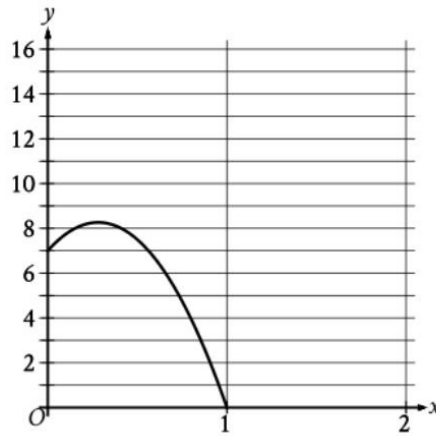
Question # ID

SAT7 M 1.6 a07ma106

Answer

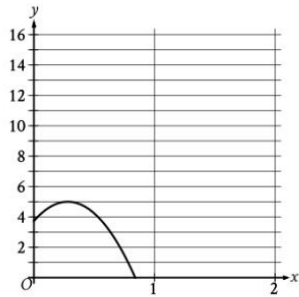
B

During the first part of an experiment, a ball was launched from a 7-foot-tall platform. The graph shows the height  $y$ , in feet, of the ball  $x$  seconds after it was launched during the first part of the experiment.

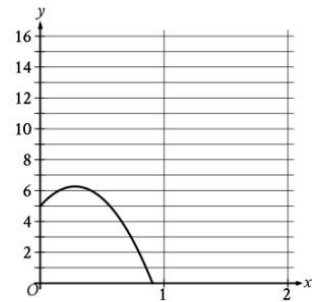


During the second part of the experiment, the ball was launched the same way, but from a platform that is 2 feet shorter than the first platform. Which of the following graphs could represent the height  $y$ , in feet, of the ball  $x$  seconds after it was launched during the second part of the experiment?

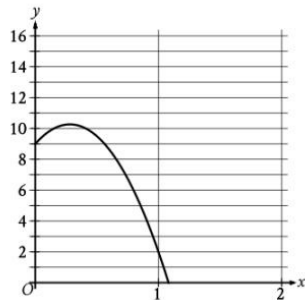
(A)



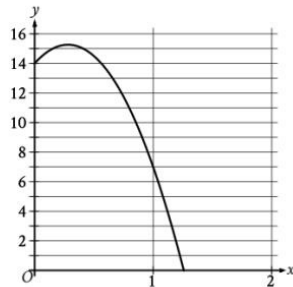
(B)



(C)



(D)



## SAT 7

## Math Module 1

Question # ID  
SAT7 M 1.7 a07ma107

Answer

90

The ratio  $x$  to  $y$  is equivalent to the ratio 9 to 5. If the value of  $x$  is 162, what is the value of  $y$ ?

SAT7 M 1.8 a07ma108

D

At a state fair, attendees can win tokens that are worth a different number of points depending on the shape. One attendee won  $S$  square tokens and  $C$  circle tokens worth a total of 1,120 points. The equation  $80S + 90C = 1,120$  represents this situation. How many more points is a circle token worth than a square token?

(A) 950

(B) 90

(C) 80

(D) 10

## SAT 7

## Math Module 1

Question # ID  
SAT7 M 1.9 a07ma109

Answer

C

$$\frac{(x+9)(x-9)}{x+9} = 7$$

What is the solution to the given equation?

(A) 7

(B) 9

(C) 16

(D) 63

SAT7 M 1.10 a07ma110

D

The equation  $12t + b = c$  relates the variables  $t$ ,  $b$ , and  $c$ . Which of the following correctly expresses the value of  $c - b$  in terms of  $t$ ?

(A)  $\frac{t}{12}$

(B)  $t$

(C)  $t + \frac{1}{12}$

(D)  $12t$

SAT7 M 1.11 a07ma111

14

What is an  $x$ -coordinate of an  $x$ -intercept of the graph of  $y = 3(x - 14)(x + 5)(x + 4)$  in the  $xy$ -plane?

## SAT 7

## Math Module 1

Question # ID  
SAT7 M 1.12 a07ma112

Answer

2.75

The function  $f$  is defined by  $f(x) = 5\left(\frac{1}{4} - x\right)^2 + \frac{11}{4}$ . What is the value of  $f\left(\frac{1}{4}\right)$ ?

SAT7 M 1.13 a07ma113

A

A circle in the  $xy$ -plane has the equation  $(x - 13)^2 + (y - k)^2 = 64$ . Which of the following gives the center of the circle and its radius?

(A) The center is at  $(13, k)$  and the radius is 8.

(B) The center is at  $(k, 13)$  and the radius is 8.

(C) The center is at  $(k, 13)$  and the radius is 64.

(D) The center is at  $(13, k)$  and the radius is 64.

SAT7 M 1.14 a07ma114

4.41

A circle has a radius of 2.1 inches. The area of the circle is  $b\pi$  square inches, where  $b$  is a constant. What is the value of  $b$ ?

# SAT 7

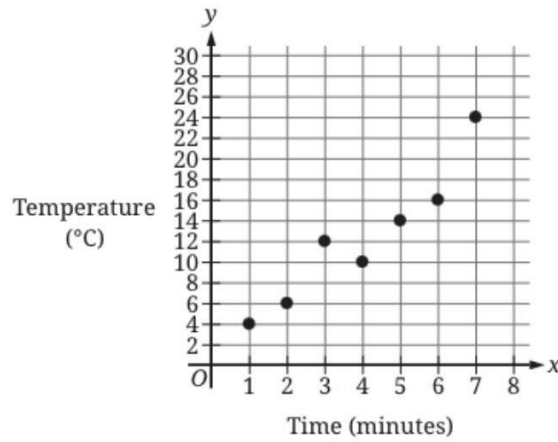
## Math Module 1

Question # ID  
SAT7 M 1.15 a07ma115

Answer

5

During a study, the temperature, in degrees Celsius ( $^{\circ}\text{C}$ ), of the air in a chamber was recorded to the nearest integer at certain times. The scatterplot shows the recorded temperature  $y$ , in  $^{\circ}\text{C}$ , of the air in the chamber  $x$  minutes after the start of the study.



What was the average rate of change, in  $^{\circ}\text{C}$  per minute, of the recorded temperature of the air in the chamber from  $x = 5$  to  $x = 7$ ?

SAT7 M 1.16 a07ma116

D

$$3y = 4x + 17$$

$$-3y = 9x - 23$$

The solution to the given system of equations is  $(x, y)$ . What is the value of  $39x$ ?

(A)  $-18$

(B)  $-6$

(C)  $6$

(D)  $18$



## SAT 7

## Math Module 1

Question # ID  
SAT7 M 1.17 a07ma117

Answer

B

The function  $f(t) = 40,000(2)^{\frac{t}{790}}$  gives the number of bacteria in a population  $t$  minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

(A) 2

(B) 790

(C) 1,580

(D) 40,000

SAT7 M 1.18 a07ma118

11

If  $5 - 7(2 - 4x) = 16 - 8(2 - 4x)$ , what is the value of  $2 - 4x$ ?

## SAT 7

## Math Module 1

Question # ID  
SAT7 M 1.19 a07ma119

Answer

A

A scientist studying the life cycle of dragonflies counted the number of dragonflies in a certain habitat each day for 46 days. On February 15, there were 99 dragonflies in the habitat. The percent increase in the number of dragonflies in the habitat from January 1 to February 15 was 12.50%. How many dragonflies were in the habitat on January 1?

(A) 88

(B) 87

(C) 12

(D) 8

SAT7 M 1.20 a07ma120

120

$$2x^2 - 8x - 7 = 0$$

One solution to the given equation can be written as  $\frac{8-\sqrt{k}}{4}$ , where  $k$  is a constant. What is the value of  $k$ ?

## SAT 7

## Math Module 1

Question # ID  
SAT7 M 1.21 a07ma121

Answer

A

The function  $f(x)$  is defined as 19 more than 4 times a number  $x$ . If  $y = f(x)$  is graphed in the  $xy$ -plane, what is the best interpretation of the  $x$ -intercept?

- (A) When  $f(x) = 0$ , the number is  $-\frac{19}{4}$ .
- (B) When the number is 0,  $f(x) = 19$ .
- (C) The value of  $f(x)$  increases by 1 for each increase of 4 in the value of the number.
- (D) For each increase of 1 in the value of the number,  $f(x)$  increases by 4.

SAT7 M 1.22 a07ma122

C

Which of the following expressions is equivalent to  $(\sin 24^\circ)(\cos 66^\circ) + (\cos 24^\circ)(\sin 66^\circ)$ ?

- (A)  $2(\cos 66^\circ)(\sin 24^\circ)$
- (B)  $2(\cos 66^\circ) + 2(\cos 24^\circ)$
- (C)  $(\cos 66^\circ)^2 + (\cos 24^\circ)^2$
- (D)  $(\cos 66^\circ)^2 + (\sin 24^\circ)^2$